Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/28/2004
PI Name:	Bhattacharjee, Subrata Ph.D.		
Project Title:	Dynamics of Flame Spread in Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92182-0001	**Congressional District:**	53
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	12/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	6
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	10
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Wakai, Kazunori (Gifu University, Japan) Takahashi, Shuhei (Gifu University, Japan)		
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	Despite intensive research on microgravity flame spread in the last decade, inspired primarily from fire safety concerns in manned space travel, a number of aspects of the flame spread problem has been identified that still remain unexplored or unresolved. Primary among those is to establish experimentally whether a spreading flame self-extinguishes in a quiescent microgravity environment, and if so, under what circumstances. Although there is some experimental evidence that for thick fuels steady spread does not occur, the phenomenon of self-extinguishment, with its importance fire-safety related consequences, deserves to be scrutinized further. Moreover, the mechanism of extinguishment remains to be addressed. For instance, we would like to answer if extinction occurs at a particular critical fuel thickness, how that thickness depends on the ambient and fuel parameters, how long does it take for a flame to extinguish, etc. To settle this important issue and address a number of related outstanding problems of flame spread, we propose four		

Task Description:	space-based experiments on flame spread over thin sheets of PMMA placed in the center of a wind tunnel in which a fully developed flow of oxygen/nitrogen mixture with prescribed average velocities can be created. The experiments are made compact by moving the fuel, spooled on two rollers, opposite to the direction of the spread to make the flame stationary. As a result, experimental parameters, especially flow velocity and fuel thickness, can be varied in a single experiment without intervention from an operator. By simply changing the location of the igniter the experiment can be re-configured for concurrent-flow spread. The project is directly related to basic understanding of flame spread important for materials selection leading to minimization of risk of ignition and fire hazard in space stations and space voyages. Besides the outstanding issues, microgravity offers a unique opportunity to link the two paradigms of flame spread - wind-aided and wind opposed configurations - by gradually reversing the direction of flow velocity in the same experiment. Moreover, the rich data set produced by the proposed experiments will help verify existing spread rate formulas and fuel development of new formulas that bring together the two regimes of concurrent-flow and opposed-flow flame spread. Currently such data is available only for flow velocities greater than the velocity induced by buoyancy driven flow. As a result flame spread formulas are unavailable or untested at the low velocity range most relevant to microgravity environment. The proposed imaging diagnostics, builds upon earlier development of emission pyrometry by the first PI, and is expected to produce improved non-obtrusive measurement of line-of-sight-averaged temperature and concentration field. The orginal goal of the proposed effort is to conduct a simple yet definitive set of experiments that, coupled with computational modeling and theoretical effort, can resolve the ten outstanding issues of flame spread. Although many of these issues involve refinement of existing models to unify different sub-regimes of flame spread, issue 3, that defines appropriate gravity level in an experiment, issues 7 and 8, that experimentally resolve the mechanism of flame extinction at quiescent environment, and issue 9, that links concurrent-flow flame spread with opposed-flow spread, are the core issues of this proposal. The simplicity of the experiment allowing a huge volume of data without any additional cost offers a unique opportunity to address the rest of the issues. The importance of fire safety in manned space travel cannot be overstated. The proposed research hopes to establish criteria under which flames, previously thought to be self propagating, extinguishes by itself. If a critical fuel thickness can be reliably predicted above which steady flames cannot be sustained over a combustible material, it will have a direct impact on how combustible materials are made fire safe in the Internal Space Station. The study of transition between wind-aided spread rate to wind-opposed spread rate will allow us to leverage the microgravity experiments to link earth bound flames, which are mostly wind-aided, to wind-opposed flames, which are theoretically better understood.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have completed the development of microgravity flame spread theory and published the results in the 30th symposium of combustion. We have also completed the construction of the SDSU Flame Tower in which a sample can be moved upward or downward at any desired velocity. Our preliminary data indicate that the flow configuraiton can be effectively controlled from wind-opposed to wind-aided spread. The object-oriented design for the 3-d, time dependent Navier-Stokes solver is complete. We have already implemented the energy equation and are now working on momentum equations
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	S. Bhattacharjee "A Time-Scale Analysis of Opposed-Flow Flame Spread" Opposed-flow flame spread over solid fuels has been extensively studied over the last three decades1-5. While most of the work have been concentrated into separate studies of the sub-categories such as the thermal regime, thick fuels, thin fuels, super-thin fuels, downward spread, kinetic regime, radiative regime, and, recently, microgravity regime, efforts to bring them together through a single analysis have been relatively few5. Although scaling arguments5,6 have been used in the past to understand the mechanism of flame spread and the complicated energy balance on the fuel surface, in this work we offer a novel time scale analysis encompassing all sub-regimes for the first time following the scale-analysis conventions established by Bejan7. Existing analytical solutions, comprehensive computational model and available experimental data will be used to validate the results and, whenever possible, determine critical values of the non-dimensional numbers arising out of scaling arguments that delineate the sub-regimes of opposed-flow flame spread. 4th Int. Symp. On Scale Modeling (ISSM-IV), Cleveland, Sept. 17-19, 2003 , Sep-2003
Abstracts for Journals and Proceedings	Paolini, C., Yeo, K.H., and Bhattacharjee, S. "An Object Oriented Formulation for the Finite Volume SIMPLER" Object oriented programming (OOP) offer several advantages over traditional structural languages. Most of the codes to implement finite-volume formulation, however, are legacy codes that are written primarily in Fortran. If a finite volume cell can be treated as an object, it can be imparted the decision making power to update its field. A cell where nothing is happening will simply turn itself off, thereby, saving unnecessary computations. Parallelization can be a natural extension of such computing approach. In this paper an object-oriented design of the SIMPLE algorithm is presented. Using Java a three-dimensional energy equation solver has been implemented. The ultimate goal of this project is to develop an object-oriented code that can be easily adopted to simulate opposed and concurrent-flow flame spread. WSS Technical Meeting of the Combustion Institute, October 22-23, UCLA, 2003 , Oct-2003
Articles in Peer-reviewed Journals	Bhattacharjee, S., Ayala, R., Wakai, K., Takahashi, S. "Opposed-Flow Flame Spread in Microgravity – Theoretical Prediction of Spread Rate and Flammability Maps" Proceedings of the Combustion Institute , Jul-2004
Articles in Peer-reviewed Journals	Subrata Bhattacharjee, Matthew D. King, Chris Paolini "Structure of Downward Spreading Flames: A Comparison of Numerical Simulation, Experimental Results and a Simplified Parabolic Theory" Combustion Theory Modeling, Vol. 7, pp. 1-17, (2003) , Dec-2003

Articles in Peer-reviewed Journals	Olson, S.L.; Hegde, U.; Bhattacharjee, S.; Deering, J.L.,; Tang, L; and Altenrich, R.A.; "Diffusive and Radiative Transport Effects on Flame Spread over Thermally Thick Solids in Microgravity" Combustion Science and Technology, Vol. 176, pp. 557-584, (2004) , Feb-2004